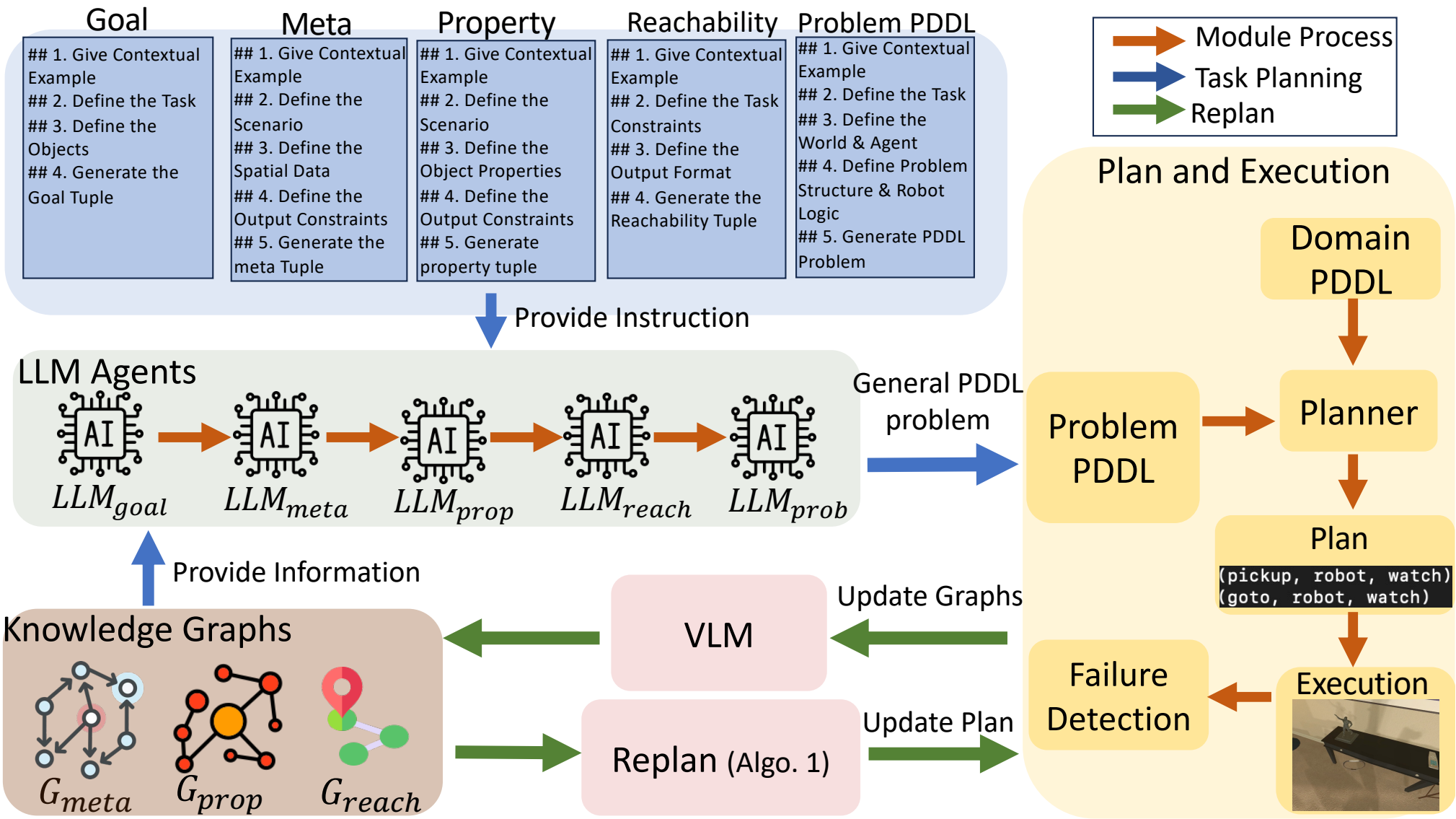
Goal
1. Give Contextual Example
2. Define the Task
3. Define the Objects
4. Generate the Goal Tuple
Meta
1. Give Contextual Example
2. Define the Scenario
3. Define the Spatial Data
4. Define the Output Constraints
5. Generate the meta Tuple
Property
1. Give Contextual Example
2. Define the Scenario
3. Define the Object Properties
4. Define the Output Constraints
5. Generate property tuple
Reachability
1. Give Contextual Example
2. Define the Task Constraints
3. Define the Output Format
4. Generate the Reachability Tuple
Problem PDDL
1. Give Contextual Example
2. Define the Task
3. Define the World & Agent
4. Define Problem Structure & Robot Logic
5. Generate PDDL Problem
Module Process
Task Planning
Replan
Provide Instruction
LLM Agents
AI
LLM_{goal}
LLM_{meta}
LLM_{prop}
LLM_{reach}
LLM_{prob}
General PDDL problem
Plan and Execution
Domain PDDL
Problem PDDL
Planner
Plan
(pickup, robot, watch)
(goto, robot, watch)
Execution
Failure Detection
Provide Information
Knowledge Graphs
G_{meta}
G_{prop}
G_{reach}
VLM
Update Graphs
Replan (Algo. 1)
Update Plan